

Comparative Evaluation of Attention-Enhanced Deep Transfer Learning Architectures: Improving Diagnostic Accuracy for Lung Cancer Detection in CT Scans

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Abstract—

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I. INTRODUCTION

Lung cancer persists as the most formidable oncological challenge globally, accounting for the highest mortality rate among all cancers, with an estimated 1.8 million deaths annually [1]. The disease's lethality is intrinsically linked to its late-stage diagnosis, where therapeutic interventions often yield limited success. Conversely, early-stage detection is associated with a significantly improved prognostic outcome, potentially increasing the five-year survival rate from less than 20

The diagnostic interpretation of CT scans, however, presents a substantial cognitive burden for radiologists. The process involves meticulously analyzing hundreds of cross-sectional slices per patient to identify nodules that may be small, low-contrast, or located in complex anatomical regions. This task is exacerbated by high inter-observer variability, perceptual fatigue, and the increasing volume of screening data, leading to potential oversights and diagnostic inconsistencies [4]. Consequently, the development of robust, automated Computer-Aided Diagnosis (CAD) systems has become a paramount research objective to serve as a synergistic second reader, enhancing diagnostic accuracy, reducing interpretation time, and mitigating human error.

The evolution of CAD systems has transitioned through distinct phases. Initial systems relied on traditional machine learning paradigms, utilizing hand-crafted features—such as texture, shape, and intensity descriptors—extracted from segmented nodules, which were then fed into classifiers like Support Vector Machines (SVMs) or random forests [5]. While pioneering, these methods were inherently limited by the quality of segmentation and the representational capacity of the engineered features, often failing to capture the complex and heterogeneous phenotypic characteristics of malignant lesions.

The advent of deep learning (DL), particularly deep convolutional neural networks (CNNs), catalyzed a paradigm shift in medical image analysis. CNNs obviate the need for manual feature engineering by autonomously learning hierarchical, multi-scale feature representations directly from raw pixel data [6]. Transfer learning—the practice of initializing models with weights pre-trained on large-scale natural image datasets like ImageNet—has proven particularly effective in the medical domain. This approach mitigates the perennial challenge of limited annotated medical data by leveraging generalized low-level feature extractors (e.g., edge and texture detectors) and focusing fine-tuning on domain-specific high-level features [7]. Architectures such as VGG16 [8], ResNet50 [9], and MobileNetV2 [10] have been extensively validated for pulmonary nodule classification, demonstrating superior performance over traditional methods.

Despite these advancements, canonical CNNs possess a fundamental limitation for medical image analysis: they employ a static, feedforward processing mechanism that treats all spatial regions within an image with equal priority. This is intrinsically misaligned with the diagnostic process, where pathological findings are often highly localized and constitute a minute fraction of the entire scan. To address this, attention mechanisms have been introduced, drawing inspiration from human visual cognition. These modules enable networks to dynamically recalibrate their focus, amplifying salient features—such as spiculated margins or ground-glass opacities indicative of malignancy—while suppressing irrelevant background information [11]. The Convolutional Block Attention Module (CBAM) [12], which sequentially infers channel-wise and spatial-wise attention maps, has shown significant efficacy in enhancing CNN performance for various medical imaging tasks.

Concurrently, a new architectural paradigm has emerged: the Vision Transformer (ViT) [13]. ViTs dispense with convolutional inductive biases entirely, instead treating an image as a sequence of patches and leveraging a self-attention mech-

anism to model global contextual relationships from the first layer. While achieving state-of-the-art results on large datasets, their performance on moderate-sized, domain-specific medical imaging datasets without extensive pre-training remains an area of active investigation [14].

While numerous studies have applied deep learning to lung nodule analysis, a rigorous comparative evaluation of an attention-enhanced CNN against a suite of modern architectures—including both foundational CNNs and a pure transformer—on a meticulously curated and balanced dataset is lacking. Furthermore, the specific integration of a computationally efficient attention module like CBAM with the robust, hierarchical feature extractor VGG16 for a three-class lung cancer diagnosis problem represents a novel contribution.

This study aims to address these gaps by conducting a comprehensive comparative analysis of state-of-the-art deep transfer learning architectures for the classification of lung CT scans into normal, benign, and malignant categories. We utilize the public IQ-OTH/NCCD dataset [15], which we have systematically pre-processed and augmented to ensure statistical balance and enhance model generalizability. The core of our work is the proposition of an attention-augmented VGG16 (AttVGG16) model, designed to synergize the deep feature extraction capabilities of VGG16 with the adaptive, region-specific focusing power of the CBAM attention mechanism.

The principal contributions of this work are fourfold:

Methodological Rigor: The systematic curation, preprocessing, and advanced augmentation of a public lung CT dataset to create a robust benchmark for model evaluation.

Comprehensive Benchmarking: An extensive empirical comparison of four leading architectures: VGG16, ResNet50, MobileNetV2, and Vision Transformer (ViT), establishing a performance baseline.

Architectural Innovation: The novel integration of a convolutional block attention module (CBAM) into the VGG16 backbone, creating the proposed AttVGG16 model for enhanced diagnostic precision.

Empirical Validation: A detailed evaluation demonstrating that the attention-enhanced architecture achieves superior diagnostic performance, conclusively underscoring the critical value of spatially adaptive feature processing in the domain of medical image analysis for lung cancer detection.

The remainder of this paper is structured as follows: Section II provides a review of relevant literature. Section III details our methodology, including the dataset, preprocessing, and architectural designs. Section IV presents the experimental results and discussion. Finally, Section V concludes the paper and suggests directions for future work.

II. LITERATURE REVIEW

The accurate and early diagnosis of lung cancer from Computed Tomography (CT) scans is a critical challenge in radiology, with deep learning (DL) emerging as a powerful tool to augment diagnostic precision. The evolution of computer-aided diagnosis (CAD) systems has transitioned from traditional machine learning relying on hand-crafted features

to deep convolutional neural networks (CNNs) capable of learning hierarchical feature representations directly from data.

Early CAD systems predominantly utilized feature extraction techniques like Scale-Invariant Feature Transform (SIFT) and Histogram of Oriented Gradients (HOG), followed by classifiers such as Support Vector Machines (SVMs) for nodule detection and classification [16]. While pioneering, these methods were often limited by their dependence on manual feature engineering, which could fail to capture the complex and subtle pathological patterns indicative of early-stage malignancies.

The advent of deep learning, particularly CNNs, marked a paradigm shift. Seminal architectures like AlexNet [17], VGGNet [18], and ResNet [19] demonstrated unprecedented performance in natural image recognition by learning multi-level abstractions. This success rapidly translated to the medical imaging domain. Studies by Setio et al. [20] and Ardila et al. [21] showcased the potential of CNNs in detecting pulmonary nodules with high sensitivity, leveraging their ability to process raw pixel data and learn discriminative features such as spiculation, lobulation, and texture. The VGG16 architecture, in particular, has been widely adopted for medical image analysis due to its deep, uniform structure, which is highly effective for feature extraction, as demonstrated in tasks ranging from retinal disease classification [22] to brain tumor segmentation [23].

However, a significant limitation of standard CNNs is their static, feedforward nature, which treats all spatial regions within an image equally. This is suboptimal for medical imaging, where discriminative features are often highly localized and occupy a small fraction of the scan. To address this, transfer learning has become a cornerstone methodology. By leveraging pre-trained models (e.g., on ImageNet) and fine-tuning them on specific medical datasets, researchers can achieve robust performance even with limited annotated data, mitigating overfitting and reducing computational costs [24]. Models like ResNet50, with its residual connections, have been favored for their ability to train very deep networks effectively [25], while MobileNetV2 offers a balance between accuracy and computational efficiency for potential deployment in resource-constrained environments [26].

More recently, attention mechanisms have been introduced to dynamically weight the importance of different features or spatial regions. Inspired by human visual attention, these mechanisms allow networks to focus on semantically salient regions. The Convolutional Block Attention Module (CBAM) [27], which sequentially infers attention maps along both channel and spatial axes, has been successfully integrated into CNN backbones to improve performance in various medical imaging tasks, including chest X-ray classification [28]. Concurrently, Vision Transformers (ViTs) [29] have emerged, challenging the dominance of CNNs by applying self-attention to sequences of image patches, thereby capturing global contextual relationships from the first layer. While promising, ViTs typically require extremely large datasets for pre-training to outperform well-established CNNs on domain-

specific tasks like medical image analysis [30].

Despite these advancements, a comparative evaluation of these advanced architectures—particularly an attention-enhanced CNN against both modern CNNs and pure transformers—for lung cancer classification on a balanced, augmented dataset remains underexplored. Most studies focus on binary classification (malignant vs. benign/normal) or use imbalanced data, which can bias model performance. Furthermore, the integration of a computationally efficient attention module like CBAM with a robust feature extractor like VGG16 for a three-class lung cancer diagnosis problem is a novel approach that warrants investigation.

This study aims to fill this gap by conducting a comprehensive comparative analysis of state-of-the-art deep transfer learning architectures, including VGG16, ResNet50, MobileNetV2, and Vision Transformer (ViT). We propose an attention-augmented VGG16 (AttVGG16) model designed to enhance feature discriminability by focusing on critical diagnostic regions within CT slices. Our work is validated on a publicly available, expertly annotated dataset that has been rigorously augmented to ensure balance and generalizability, providing a robust benchmark for lung cancer diagnostic accuracy.

Fig. 1:

III. METHODS AND MATERIALS

A. Data Collection & Preprocessing

This study employs the Iraq-Oncology Teaching Hospital/National Center for Cancer Diseases (IQ-OTH/NCCD) lung cancer dataset [1], a publicly available collection of thoracic computed tomography (CT) scans. The dataset was curated over a three-month period in the fall of 2019 at two specialized Iraqi cancer centers, namely the Oncology Teaching Hospital and the National Center for Cancer Diseases. It comprises CT scans from 110 subjects, including patients diagnosed with lung cancer at varying stages as well as healthy controls. Expert oncologists and radiologists from these centers carefully annotated the dataset, ensuring high-quality clinical labels.

The original dataset consists of 1,190 CT slices, derived from 40 malignant cases, 15 benign cases, and 55 normal cases. Each patient contributed between 80 and 200 slices, acquired using a Siemens SOMATOM scanner under the following imaging protocol: tube voltage of 120 kV, slice thickness of 1 mm, window width ranging from 350 to 1200 Hounsfield Units (HU), and window center between 50 and 600 HU. All scans were performed with subjects under full-inspiration breath-hold to minimize motion artifacts. The raw data was originally stored in Digital Imaging and Communications in Medicine (DICOM) format, which was subsequently de-identified in compliance with ethical and privacy regulations. Institutional Review Board (IRB) approval was obtained from the participating medical centers, while written consent was waived by the oversight review board [1].

To address the inherent class imbalance (40 malignant, 15 benign, and 55 normal cases), we extended the dataset through

systematic data augmentation until each class reached 600 samples, yielding a balanced dataset of 1,800 images. The augmentation pipeline employed a combination of geometric and photometric transformations, including horizontal and vertical flips, left-right and top-bottom rotations, scaling, random cropping, translation, and mild intensity variations. These transformations were carefully selected to preserve the diagnostic integrity of lung structures while increasing intra-class diversity and enhancing the generalization capacity of downstream models [2,3].

B. Best classifier selection

1) *Proposed Method*: The proposed approach introduces an attention-based VGG16 transfer learning framework (AttVGG16) to evaluate the effectiveness of deep feature representations in classifying CT scan images of lung cancer. A modern attention mechanism enhances the canonical VGG16 architecture by enabling the model to focus on the most discriminative regions of the input images, thereby improving classification performance. The attention mechanism dynamically assigns weights to salient spatial features, allowing the network to overcome limitations of fixed receptive fields and extract more informative representations from complex medical images.

Initially, we employed several state-of-the-art convolutional neural networks, including VGG16, ResNet50, MobileNetV2, and Vision Transformer (ViT), for baseline performance comparison. Among these, VGG16 demonstrated strong feature extraction capabilities due to its deep hierarchical structure, making it suitable for augmentation with an attention mechanism.

In the proposed framework, CT scan images are fed into the VGG16 backbone to extract hierarchical feature maps. The attention module then emphasizes regions that are most indicative of malignancy or benign abnormalities. The attention-weighted feature maps are subsequently passed through fully connected layers to produce the final classification into normal, benign, or malignant categories.

By integrating attention into VGG16, the proposed AttVGG16 architecture mitigates the loss of spatial information in high-level feature maps and enhances the model's ability to detect subtle differences between classes. This framework is particularly advantageous for medical image analysis, where discriminative patterns may be localized and sparse.

2) *Vgg16*: For the image classification task, we employed the VGG16 architecture, a deep convolutional neural network introduced by Simonyan and Zisserman (2014)[1]. The model consists of 16 layers, including 13 convolutional layers and 3 fully connected layers, with small 3x3 convolution filters and 2x2 max-pooling layers to reduce spatial dimensions (Simonyan Zisserman, 2014)[1]. The network utilizes ReLU activation functions, and the final layer uses a softmax activation to output class probabilities. Training was performed using Stochastic Gradient Descent (SGD) with momentum, as described by Sutskever et al. (2013)[2], optimizing the cate-

gorical cross-entropy loss function. The convolution operation is defined as:

$$f_{\text{out}} = \sigma(\mathbf{W} * f_{\text{in}} + \mathbf{b})$$

where f_{out} is the output feature map, $\mathbf{W}$ is the convolution kernel, f_{in} is the input feature map, $\mathbf{b}$ is the bias term, and σ is the ReLU activation function. The pooling operation is a standard 2×2 max-pooling layer with a stride of 2, effectively down-sampling the spatial dimensions. For the fully connected layers, the output of the final convolutional layer is flattened and passed through three dense layers, each followed by a ReLU activation function. The final layer uses softmax activation to output a probability distribution over C classes:

$$\hat{y}_i = \frac{e^{z_i}}{\sum_{j=1}^C e^{z_j}}$$

where $\hat{y}_i$ is the predicted probability for class i , and z_i is the raw output (logit) for class i .

Training was performed using Stochastic Gradient Descent (SGD) with momentum μ , updating the weights according to:

$$\mathbf{w}_{t+1} = \mathbf{w}_t - \eta \nabla L(\mathbf{w}_t) + \mu(\mathbf{w}_t - \mathbf{w}_{t-1})$$

where $\mathbf{w}_t$ is the weight vector at iteration t , η is the learning rate, and $\nabla L(\mathbf{w}_t)$ is the gradient of the categorical cross-entropy loss function L :

$$L(\mathbf{y}, \hat{\mathbf{y}}) = - \sum_{i=1}^C y_i \log(\hat{y}_i)$$

where $\mathbf{y}$ is the ground truth label vector and $\hat{\mathbf{y}}$ is the predicted probability vector.

To prevent overfitting, data augmentation techniques (Krizhevsky et al., 2012)[3] were applied, and pre-trained ImageNet weights were used for transfer learning (Donahue et al., 2014)[4]. Early stopping was employed to halt training when validation accuracy plateaued, and the model was implemented in Keras with TensorFlow as the backend. The VGG16 feature extraction model has the capability of extracting a huge amount of data and results in good accuracy. It is one of the most popular techniques of image feature extraction, which performs better if any DL model is applied for classification tasks. [5]

3) *Resnet50*: For the image classification task, we utilized the ResNet50 architecture, a deep residual learning framework introduced by He et al. (2015) [6]. Unlike conventional CNNs such as VGG16, which learn a direct mapping from inputs to outputs, ResNet employs residual connections to address the degradation problem in very deep networks. This design enables effective gradient backpropagation through identity shortcuts, allowing the training of networks with over 100 layers. Residual Learning In a residual block, instead of approximating the desired mapping $\mathcal{H}(\mathbf{x})$ directly, the network learns a residual function $\mathcal{F}(\mathbf{x}, \{\mathbf{W}_i\})$ such that:

$$\mathbf{y} = \mathcal{F}(\mathbf{x}, \{\mathbf{W}_i\}) + \mathbf{x}$$

where $\mathbf{x}$ denotes the input feature map, $\mathbf{y}$ is the block output, and $\{\mathbf{W}_i\}$ represents the convolutional kernels.

The addition of the input $\mathbf{x}$ through an identity skip connection ensures that the block can preserve low-level features and propagate gradients effectively. The residual mapping $\mathcal{F}$ typically consists of two or three stacked convolutional layers with batch normalization and ReLU activation functions.

Convolution and Activation Each convolutional operation in ResNet is defined as:

$$\mathbf{f}_{\text{out}} = \sigma(\mathbf{W} * \mathbf{f}_{\text{in}} + \mathbf{b})$$

where $\mathbf{f}_{\text{in}}$ and $\mathbf{f}_{\text{out}}$ are the input and output feature maps, $\mathbf{W}$ is the convolution kernel, $\mathbf{b}$ the bias, and σ the ReLU non-linearity. Spatial dimensionality reduction is performed via strided convolutions and global average pooling, replacing the large fully connected layers used in VGG16 (Simonyan & Zisserman, 2014) [?].

Output Layer and Training The final dense layer applies the softmax activation to generate class probabilities:

$$\hat{y}_i = \frac{e^{z_i}}{\sum_{j=1}^C e^{z_j}}$$

where z_i is the logit for class i among C classes.

The model was trained using Stochastic Gradient Descent with momentum [?], optimizing the categorical cross-entropy loss:

$$L(\mathbf{y}, \hat{\mathbf{y}}) = - \sum_{i=1}^C y_i \log(\hat{y}_i)$$

where $\mathbf{y}$ is the ground truth label vector and $\hat{\mathbf{y}}$ is the predicted probability vector.

ResNet50's residual connections make it one of the most reliable architectures for deep feature extraction and classification tasks. Recent studies have confirmed its superior performance in various biomedical imaging applications compared to conventional CNNs such as VGG16 (Sharma et al., 2022) [9].

4) *MobileNet V2*: For the image classification task, we also employed the MobileNetV2 architecture, introduced by Sandler et al. (2018) [10]. Unlike conventional deep CNNs such as VGG16 [2] and ResNet50 [3], MobileNetV2 is specifically designed for computational efficiency on mobile and embedded devices while maintaining competitive accuracy. The network achieves this by combining depthwise separable convolutions, inverted residuals, and linear bottlenecks. Depthwise Separable Convolution Standard convolution simultaneously filters and combines feature maps, which is computationally expensive. MobileNetV2 factorizes this into depthwise convolution followed by pointwise convolution (1×1).

Formally, given an input tensor of size $D_F \times D_F \times M$ and producing an output with N channels, a standard convolution requires:

$$\text{Cost}_{\text{standard}} = D_K^2 \cdot M \cdot N \cdot D_F^2$$

where D_K is the kernel size.

Depthwise separable convolution reduces the computation to:

$$\text{Cost}_{\text{depthwise}} = D_K^2 \cdot M \cdot D_F^2 + M \cdot N \cdot D_F^2$$

This decomposition significantly reduces parameters and floating-point operations (FLOPs), making the model efficient for real-time applications. Inverted Residual and Linear Bottleneck The core building block of MobileNetV2 is the inverted residual structure with a linear bottleneck. Instead of using wide input–narrow output blocks (as in ResNet), MobileNetV2 expands the input with a pointwise convolution, applies a depthwise convolution, and then projects back to a low-dimensional bottleneck:

$$\mathbf{y} = \mathbf{W}_p \sigma(\mathbf{W}_d \sigma(\mathbf{W}_e \mathbf{x}))$$

where:

- $\mathbf{x}$: input feature map
- $\mathbf{W}_e$: expansion (1×1 convolution, increasing dimensionality)
- $\mathbf{W}_d$: depthwise convolution
- $\mathbf{W}_p$: projection (1×1 convolution, reducing dimensionality)
- σ : ReLU6 activation for non-linearity

The shortcut connection is applied only when input and output dimensions match:

$$\mathbf{y} = \mathcal{F}(\mathbf{x}) + \mathbf{x}$$

This inverted residual design preserves representational power while keeping computations low. The linear bottleneck avoids non-linear activations at the projection step, preventing loss of information in low-dimensional feature spaces.

Activation and Output Layer MobileNetV2 employs the ReLU6 activation function, defined as:

$$\sigma(z) = \min(\max(0, z), 6)$$

which improves robustness for low-precision computations.

The final fully connected layer applies softmax activation to produce class probabilities:

$$\hat{y}_i = \frac{e^{z_i}}{\sum_{j=1}^C e^{z_j}}$$

where z_i is the logit for class i among C classes.

Training The network was trained using Stochastic Gradient Descent (SGD) with momentum [?] and the categorical cross-entropy loss:

$$L(\mathbf{y}, \hat{\mathbf{y}}) = - \sum_{i=1}^C y_i \log(\hat{y}_i)$$

where $\mathbf{y}$ is the ground truth label vector and $\hat{\mathbf{y}}$ is the predicted probability vector.

5) *Vision Transformer (ViT)*: For the image classification task, we implemented the Vision Transformer (ViT) architecture, a novel approach that dispenses with convolutional inductive biases entirely and applies a pure Transformer model, originally designed for natural language processing, directly to sequences of image patches (Dosovitskiy et al., 2020)[11]. The core premise is to treat an image not as a spatially hierarchical structure but as a sequence of tokens, leveraging the self-attention mechanism to model global dependencies across all patches from the first layer.

An input image $\mathbf{x} \in \mathbb{R}^{H \times W \times C}$ is partitioned into a sequence of N non-overlapping patches $\mathbf{x}_p \in \mathbb{R}^{N \times (P^2 \cdot C)}$, where (P, P) is the resolution of each patch and $N = \frac{HW}{P^2}$ is the resulting sequence length.

Each patch is then flattened and linearly projected into a D -dimensional embedding space using a trainable matrix $\mathbf{E} \in \mathbb{R}^{(P^2 \cdot C) \times D}$ [?].

To retain crucial positional information of the patches, a learnable positional embedding $\mathbf{E}_{pos} \in \mathbb{R}^{(N+1) \times D}$ is added to the patch embeddings. A learnable [class] token embedding $\mathbf{z}_0^0 \in \mathbb{R}^D$ is prepended to the sequence, whose state at the output of the Transformer encoder serves as the global image representation $\mathbf{y}$ [?].

The input to the Transformer encoder is thus:

$$\mathbf{z}_0 = [\mathbf{x}_{\text{class}}; \mathbf{x}_{p1}\mathbf{E}; \mathbf{x}_{p2}\mathbf{E}; \dots; \mathbf{x}_{pN}\mathbf{E}] + \mathbf{E}_{pos}$$

Transformer Encoder: The Transformer encoder consists of L identical layers. Each layer l comprises a Multi-Head Self-Attention (MSA) block and a Multi-Layer Perceptron (MLP) block, with LayerNorm (LN) applied before each block and residual connections after every block (Vaswani et al., 2017)[13]:

$$\mathbf{z}'_l = \text{MSA}(\text{LN}(\mathbf{z}_{l-1})) + \mathbf{z}_{l-1}$$

$$\mathbf{z}_l = \text{MLP}(\text{LN}(\mathbf{z}'_l)) + \mathbf{z}'_l$$

The Multi-Head Self-Attention (MSA) mechanism enables the model to jointly attend to information from different representation subspaces at different positions. For a single self-attention head, the input sequence $\mathbf{z}$ is transformed into Query ($\mathbf{Q}$), Key ($\mathbf{K}$), and Value ($\mathbf{V}$) matrices via learnable linear projections. The self-attention is computed as:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right) \mathbf{V}$$

where d_k is the dimensionality of the keys, and the scaling factor $\sqrt{d_k}$ stabilizes gradients during training [?].

The outputs of h attention heads are concatenated and linearly projected to form the final MSA output. The MLP block contains two linear layers with a GELU non-linearity [?] in between:

$$\text{MLP}(\mathbf{z}) = \text{Linear}(\text{GELU}(\text{Linear}(\mathbf{z})))$$

Classification Head: The state of the [class] token $\mathbf{z}_L^0$ at the output of the L -th encoder layer is passed through a LayerNorm layer and fed into a single linear layer (the MLP head) that outputs the logits $\mathbf{z}$ for each class.

The final class probabilities are obtained using the softmax activation function:

$$\hat{y}_i = \frac{e^{z_i}}{\sum_{j=1}^C e^{z_j}}$$

where C is the total number of classes.

Training: The model was trained end-to-end by minimizing the categorical cross-entropy loss using the AdamW optimizer (Loshchilov Hutter, 2017)[15], which decouples weight decay from the gradient update, providing superior performance and regularization for transformer-based models. Extensive data augmentation and regularization strategies, including RandAugment (Cubuk et al., 2020) and dropout within the MLP blocks, were critical to achieving high performance on mid-sized datasets, compensating for the lack of inherent convolutional inductive biases.

$$L(\mathbf{y}, \hat{\mathbf{y}}) = - \sum_{i=1}^C y_i \log(\hat{y}_i)$$

6) *Attention-Based VGG16:* To enhance feature extraction for lung cancer detection from CT images, we employed an attention-augmented VGG16 architecture, which integrates convolutional feature extraction with attention mechanisms for adaptive focus on salient regions. While the baseline VGG16 model (Simonyan Zisserman, 2014) [1] captures hierarchical spatial features through stacked convolutional layers, the attention module was incorporated after the final convolutional block to strengthen discriminative representation by re-weighting channel and spatial information.

Attention Mechanism: We utilized a Convolutional Block Attention Module (CBAM) (Woo et al., 2018) [2], which sequentially applies channel attention and spatial attention to the extracted feature maps.

For an intermediate feature map $\mathbf{F} \in \mathbb{R}^{H \times W \times C}$, channel attention is formulated as:

$$M_c(\mathbf{F}) = \sigma(\text{MLP}(\text{AvgPool}(\mathbf{F})) + \text{MLP}(\text{MaxPool}(\mathbf{F})))$$

where $M_c(\mathbf{F}) \in \mathbb{R}^{1 \times 1 \times C}$ denotes the channel attention map, and σ is the sigmoid activation.

The input feature map is then re-weighted by:

$$\mathbf{F}' = M_c(\mathbf{F}) \otimes \mathbf{F}$$

where $\otimes$ represents element-wise multiplication.

Next, spatial attention is computed by applying a convolutional layer over concatenated average- and max-pooled channel descriptors:

$$M_s(\mathbf{F}') = \sigma(f^{7 \times 7}([\text{AvgPool}(\mathbf{F}'); \text{MaxPool}(\mathbf{F}')]))$$

where $f^{7 \times 7}$ denotes a convolution with a 7×7 kernel, yielding a spatial attention map $M_s(\mathbf{F}') \in \mathbb{R}^{H \times W \times 1}$.

The refined feature representation is then obtained as:

$$\mathbf{F}'' = M_s(\mathbf{F}') \otimes \mathbf{F}'$$

This attention-enhanced feature map is subsequently passed to the fully connected layers of VGG16 for classification.

Training Strategy: The model was initialized with ImageNet pre-trained weights and fine-tuned on the lung CT dataset, employing Stochastic Gradient Descent (SGD) with momentum [?]. The categorical cross-entropy loss was minimized:

$$L(\mathbf{y}, \hat{\mathbf{y}}) = - \sum_{i=1}^C y_i \log(\hat{y}_i)$$

where $\mathbf{y}$ and $\hat{\mathbf{y}}$ represent ground-truth and predicted class distributions, respectively.

To mitigate overfitting, extensive data augmentation (random rotations, horizontal flips, and intensity normalization) was employed, along with early stopping based on validation loss.

The incorporation of attention enables the network to prioritize diagnostically relevant lung regions, improving the model's ability to distinguish malignant from benign patterns in CT scans.

C. Performance evaluation metrics

Success rates in classification algorithms may be measured using a variety of performance metrics. To measure how well our technique predicts outcomes, we employed the following five metrics in this paper.

$$\text{Precision}(Pre) = \frac{TP}{TP + FP} \quad (1)$$

$$\text{Accuracy}(Acc) = \frac{TP + TN}{TP + TN + FP + FN} \quad (2)$$

$$\text{Sensitivity}(Sen) = \frac{TP}{TP + FN} \quad (3)$$

$$\text{Specificity}(Spe) = \frac{TN}{TN + FP} \quad (4)$$

$$MCC = \frac{TP * TN - FP * FN}{\sqrt{(TP + FP)(TN + FP)(TN + FN)(TP + FN)}} \quad (5)$$

$$F1 - Score = \frac{2 * (Recall * Precision)}{Recall + Precision} \quad (6)$$

Where TP, FP, TN, and FN denote true positive, false positive, true negative false negative respectively. The Receiver Operating Characteristic (ROC) curve is a graph that compares the true positive rate (TPR) to the false positive rate (FPR) for a number of various threshold values. TPR refers to sensitivity, whereas FPR refers to 1-specificity.

IV. RESULTS

V. DISCUSSIONS

VI. CONCLUSION

VII. ACKNOWLEDGMENT

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[1]